

Dear Editor,

We wish to submit our manuscript, “Probabilistic reconstruction of charge stability diagrams from sparse ray measurements”, for consideration as a Regular Paper in the Japanese Journal of Applied Physics.

Tuning a semiconductor quantum dot array requires the charge stability diagram, and acquiring that diagram by a dense two-dimensional voltage scan is the step that does not scale. In this work we show that the charge-transition lines of a double quantum dot can be recovered from a small fraction of the gate-voltage plane measured along one-dimensional rays. A fan of 8 rays sampled at 60 points touches 4.7 % of the plane; a U-Net is given only those measurements, as two image channels, and returns one probability per pixel that a transition line passes through it. On 50 held-out devices the recovered lines reach $F1 = 0.817$ at a tolerance of one pixel.

We believe the work suits this journal for two reasons. It continues a line we published here previously on ray-based tuning of a GaAs quadruple-dot array [Jpn. J. Appl. Phys. 64, 02SP22 (2025)], moving from classifying a regime to reconstructing the diagram itself. And its concern is the measurement cost of device tuning, which is a practical obstacle to scaling quantum dot arrays rather than a purely algorithmic question.

We have tried to report the result in a form that cannot be over-read. The measurement budget is quoted as the fraction of the plane actually measured. The threshold that turns the probability map into lines is fitted on a validation split carved out of the training devices and never on the held-out set. Because transition lines are one pixel wide, the score is a tolerant F1 and is always given with its tolerance and with the area that tolerance covers. We also state plainly that the devices are simulated and the simulation is noise-free, so the claim established here concerns recoverability of the geometry and not yet validation against measured data.

The manuscript is original, has not been published elsewhere, and is not under consideration by another journal. All authors have read and approved the submission and declare no conflict of interest. The code and the analysis that produced every figure and number in the manuscript are available in a public repository, whose address is given in the paper.

Thank you for considering our work.

Yours sincerely,

Ehsan Alizadeh Kashtiban, on behalf of all authors

Department of Physics, Graduate School of Science, Osaka University

E-mail: alizadeh21@sanken.osaka-u.ac.jp